

Centered Natural-Parameter Group Selection for Posterior-Score Contrast Support in von Mises–Fisher Mixtures

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Abstract

Finite mixtures of von Mises–Fisher distributions are widely used for directional clustering, but existing sparse formulations typically target component prototypes rather than coordinates that separate posterior scores. We define posterior-score contrast support through pairwise posterior log-score contrasts and express it using centered natural parameters. This leads to the Eta-centered group lasso (E-CGL), which applies a coordinate-wise group penalty to centered contrasts while leaving common natural-parameter effects unpenalized; E-ACGL is considered as an adaptive extension. Estimation combines a guarded proximal-gradient generalized-EM path with target-preserving support-constrained refitting and information-criterion selection conditional on the number of components. We establish the centered decomposition, label-invariant support, the closed-form proximal map, and monotonicity of accepted updates. Oracle-error-calibrated simulations show accurate sparse-support recovery at larger sample sizes and identify finite-sample limitations under dense or weak support. Across five locked Classic3 holdouts, E-CGL retained 1,343.0 of 2,000 coordinates on average; relative to dense free- κ_k vMF, its mean ARI difference was -0.0036 and its NLL increase was 0.6705 per document. In BBCSport, every sparse method had higher held-out NLL than its matched dense model in all five splits. The method is intended for sparse or compressible posterior-score contrasts rather than as a general replacement for dense or prototype-oriented mixtures.

Keywords: Directional data; mixture model; von Mises-Fisher distribution;

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1. Introduction

Directional representations arise whenever only the orientation of an observation is meaningful. Common examples include normalized document vectors, compositional profiles after normalization, and embedding vectors on the unit hypersphere. The von Mises-Fisher (vMF) distribution is a standard model for such data, and finite vMF mixtures provide a model-based alternative to spherical k -means (Banerjee et al., 2005). In high dimensions, however, the fitted component structure can involve many coordinates, making both estimation and interpretation difficult.

Sparse vMF mixtures have primarily been formulated through sparsity of the component direction vectors. Rossi and Barbaro (2022), for example, penalize the entries of μ_k to estimate sparse directional prototypes. This target is appropriate when the scientific question concerns which coordinates describe each component prototype. It is not identical to asking which coordinates distinguish components in posterior classification. A coordinate can be active in every prototype and still contribute almost equally to all component scores. Conversely, a coordinate with a modest directional loading can be influential when component concentrations differ. The distinction is therefore between prototype support and the feature-dependent linear support of posterior component comparisons.

This paper targets the latter support through the natural parameters $\eta_k = \kappa_k \mu_k$. For each observed coordinate, the component-specific natural parameters are decomposed into a common baseline and a centered contrast. The proposed E-CGL penalty groups the centered contrast across components, so its unit of regularization is the same coordinate-level unit used to define the target support. The common baseline remains unpenalized. E-ACGL is considered as an adaptive extension with weights fixed from a dense initial fit.

The construction draws on contrast penalties and grouped variable selection, but applies them to a different estimand. Penalized Gaussian mixtures have used entry-wise mean shrinkage, grouped component means, and pairwise mean fusion (Pan and Shen, 2007; Xie et al., 2008; Guo et al., 2010). Sum-to-zero factor contrasts have also been used to separate common levels from heterogeneity (Bondell and Reich, 2009), while other directional-mixture methods target feature saliency, block structure, or hierarchical

parameter regularization (Amayri and Bouguila, 2013; Salah et al., 2016; Gopal and Yang, 2014; Taghia et al., 2014; Salah and Nadif, 2019). Discriminative subspace approaches provide another route to variable relevance in model-based clustering (Bouveyron and Brunet, 2012; Bouveyron and Brunet-Saumard, 2014). The contribution here is the alignment of a centered natural-parameter estimand, a coordinate-group penalty, and posterior linear-score interpretation in a vMF mixture.

Candidate supports are generated by a guarded path with proximal gradient updates inside generalized EM. They are compared after support-constrained refitting that preserves the target. The primary selector is a practical BIC computed from the refitted likelihood; alternative degrees of freedom and EBIC rules are treated as sensitivity analyses. Support-recovery claims condition on fixed K . When K is unknown, component-number selection and support selection are examined in separate stages rather than through a single joint criterion.

The empirical study has three parts. Oracle-error calibrated simulations compare posterior-score support recovery across sample size, ambient dimension, and concentration heterogeneity. Shared-background and dense-support designs separate the proposed estimand from prototype sparsity and record settings in which sparse posterior-score support is inappropriate. Locked repeated-holdout analyses of Classic3 and BBCSport examine clustering, coordinate reduction, and token-level contrasts on vocabulary-aligned directional representations.

The contributions are therefore: (i) a posterior linear-score support estimand for vMF mixtures and its centered representation; (ii) E-CGL and its adaptive extension, together with a guarded estimation and exact-refit procedure; and (iii) an empirical evaluation that distinguishes the intended sparse-support regime from prototype-oriented and dense-support settings. E-CGL is not intended to replace sparse-prototype methods when prototype sparsity is the scientific target.

The remainder of the paper is organized as follows. Section 2 presents the vMF model, posterior-score contrast support, and centered group regularization. Section 3 describes the estimation algorithm, pathwise selection and refitting, and the structural and algorithmic properties used in the analysis. Sections 4 and 5 report the numerical studies and locked real-data analyses, respectively. Section 6 concludes with the empirical scope and limitations of the method.

2. Model and centered regularization

2.1. vMF mixture in natural parameters

Let $x_i \in \mathbb{S}^{d-1}$, $i = 1, \dots, n$, denote unit-normalized directional observations. A K -component vMF mixture has density

$$f(x_i; \Theta) = \sum_{k=1}^K \pi_k C_d(\kappa_k) \exp\{\kappa_k \mu_k^\top x_i\},$$

where $\pi_k > 0$, $\sum_k \pi_k = 1$, $\|\mu_k\|_2 = 1$, and $\kappa_k \geq 0$. Its normalizing constant is

$$C_d(\kappa) = \frac{\kappa^{d/2-1}}{(2\pi)^{d/2} I_{d/2-1}(\kappa)}.$$

At $\kappa = 0$, this expression is interpreted by continuous extension as $C_d(0) = \Gamma(d/2)/(2\pi^{d/2})$.

Writing $\eta_k = \kappa_k \mu_k \in \mathbb{R}^d$ gives

$$f(x_i; \Theta) = \sum_{k=1}^K \pi_k C_d(\|\eta_k\|_2) \exp(\eta_k^\top x_i).$$

The component-level mapping back to (μ_k, κ_k) and its zero-concentration exception are recorded in Section 3.4. The observed log-likelihood is

$$\ell(\Theta) = \sum_{i=1}^n \log \left\{ \sum_{k=1}^K \pi_k C_d(\|\eta_k\|_2) \exp(\eta_k^\top x_i) \right\},$$

2.2. Posterior-score contrast support

The component score

$$s_k(x) = \log \pi_k + \log C_d(\|\eta_k\|_2) + \eta_k^\top x$$

implies the pairwise posterior log-odds representation

$$\log \frac{\Pr(Z = k \mid x)}{\Pr(Z = \ell \mid x)} = a_{k\ell} + (\eta_k - \eta_\ell)^\top x,$$

$$a_{k\ell} = \log \frac{\pi_k}{\pi_\ell} + \log \frac{C_d(\|\eta_k\|_2)}{C_d(\|\eta_\ell\|_2)}.$$

Thus $\eta_k - \eta_\ell$, rather than $\mu_k - \mu_\ell$, is the coefficient of the feature-dependent linear term. For coordinate j , let

$$H_K = I_K - K^{-1} \mathbf{1}_K \mathbf{1}_K^\top, \quad c_{\cdot j} = H_K \eta_{\cdot j} = \eta_{\cdot j} - \bar{\eta}_j \mathbf{1}_K,$$

where $\bar{\eta}_j = K^{-1} \sum_k \eta_{kj}$. We define

$$S_\eta = \{j : \|c_{\cdot j}\|_2 > 0\} = \bigcup_{k < \ell} \{j : \eta_{kj} - \eta_{\ell j} \neq 0\}.$$

We use *posterior-score contrast support* for this feature-dependent linear support. It does not include the intercepts $a_{k\ell}$. A coordinate with $c_{\cdot j} = 0$ cancels from every pairwise linear term, although its common baseline $\bar{\eta}_j$ remains in the fitted component densities and may affect $a_{k\ell}$ through the concentration norms. Proposition 1 gives the unique centered decomposition and its pairwise-dispersion identity.

Accordingly, $c_{\cdot j} \neq 0$ defines a decision coordinate; $c_{\cdot j} = 0$ with $\bar{\eta}_j \neq 0$ defines a common coordinate; and $c_{\cdot j} = 0$ with $\bar{\eta}_j = 0$ defines a globally null coordinate. These categories distinguish the proposed target from prototype support, which records nonzero component coordinates without requiring a difference between components.

The support is invariant to permutations of component labels because such a permutation only reorders the entries of $c_{\cdot j}$. It is not invariant to an arbitrary orthogonal rotation of the feature axes. As with other coordinate-selection procedures, its interpretation is therefore relative to the chosen coordinate system.

2.3. Centered group regularization (E-CGL)

For fixed K , the primary penalty is

$$P_{\text{CGL}}(\eta) = \lambda_\eta \sum_{j=1}^d \|H_K \eta_{\cdot j}\|_2 = \lambda_\eta \sum_{j=1}^d \|c_{\cdot j}\|_2.$$

Each group is one observed coordinate across the K component contrasts, so the unit of regularization matches the definition of S_η . The common baseline lies in the null space of H_K and is not penalized. The first expression writes the penalty as a function of η , whereas the second displays its centered-contrast target. This coordinate-group construction follows the grouped-variable regularization principle of Yuan and Lin (2006). By contrast,

$$\|\eta_{\cdot j}\|_2^2 = K\bar{\eta}_j^2 + \|c_{\cdot j}\|_2^2,$$

so a raw- η group penalty combines common and contrast effects. A direction-space penalty targets prototype coordinates and, when the κ_k differ, does not directly target the coefficients $\eta_k - \eta_\ell$ in the posterior log-odds. These are distinct estimands, not competing parameterizations of the same support.

An entry-wise centered penalty selects component-coordinate entries separately: $\lambda_\eta \sum_{k,j} |c_{kj}|$. E-CGL instead selects the complete coordinate contrast as one group. The resulting proximal map and label invariance are stated in Propositions 3 and 2.

2.4. Adaptive extension

E-ACGL uses

$$P_{\text{ACGL}}(\eta) = \lambda_\eta \sum_{j=1}^d w_j \|H_K \eta_{\cdot j}\|_2 = \lambda_\eta \sum_{j=1}^d w_j \|c_{\cdot j}\|_2,$$

with weights fixed from the dense initial fit:

$$w_j^{\text{raw}} = (\|c_{\cdot j}^{\text{init}}\|_2 + \epsilon)^{-\gamma}, \quad w_j = \frac{w_j^{\text{raw}}}{\text{median}_{1 \leq h \leq d} w_h^{\text{raw}}}.$$

The numerical analyses use $\gamma = 1$ and $\epsilon = 10^{-6}$, and the weights remain fixed throughout optimization and path selection. E-CGL, with $w_j = 1$, is the primary estimator; E-ACGL is an adaptive extension and no oracle-property claim is made. This weighting follows the adaptive regularization principle of Zou (2006), applied here to centered coordinate groups.

3. Estimation and properties

3.1. Penalized objective and component updates

For fixed K and λ_η , the path-generating criterion is

$$\begin{aligned} \mathcal{L}_{\lambda_\eta}(\Theta) &= \ell(\Theta) - \lambda_\eta \sum_{j=1}^d w_j \|H_K \eta_{\cdot j}\|_2 \\ &= \ell(\Theta) - \lambda_\eta \sum_{j=1}^d w_j \|c_{\cdot j}\|_2. \end{aligned} \tag{1}$$

where $w_j = 1$ for E-CGL and the E-ACGL weights are fixed from the dense initial fit. The penalty and the vMF normalizing constant are coupled through η_k , so the natural-parameter update is computed by a guarded proximal-gradient generalized M-step.

3.1.1. E-step and fixed-responsibility subproblem

At outer iteration t , compute $\tau_{ik}^{(t)}$ and

$$\tau_{ik}^{(t)} = \frac{\pi_k^{(t)} C_d(\|\eta_k^{(t)}\|_2) \exp\{(\eta_k^{(t)})^\top x_i\}}{\sum_{h=1}^K \pi_h^{(t)} C_d(\|\eta_h^{(t)}\|_2) \exp\{(\eta_h^{(t)})^\top x_i\}}, \quad (2)$$

and define

$$N_k^{(t)} = \sum_{i=1}^n \tau_{ik}^{(t)}, \quad r_k^{(t)} = \sum_{i=1}^n \tau_{ik}^{(t)} x_i.$$

Conditional on these responsibilities, the natural-parameter block minimizes

$$F_t(\eta) = f_t(\eta) + g(\eta),$$

$$f_t(\eta) = - \sum_{k=1}^K \left\{ (r_k^{(t)})^\top \eta_k + N_k^{(t)} \log C_d(\|\eta_k\|_2) \right\},$$

$$g(\eta) = \lambda_\eta \sum_{j=1}^d w_j \|H_K \eta_j\|_2.$$

The smooth term is convex because $-\log C_d(\|\eta_k\|_2)$ is the vMF log-partition function. Its gradient is

$$\nabla_{\eta_k} f_t(\eta) = N_k^{(t)} A_d(\|\eta_k\|_2) \frac{\eta_k}{\|\eta_k\|_2} - r_k^{(t)},$$

with the continuous value $-r_k^{(t)}$ at $\eta_k = 0$, where A_d is the vMF mean-resultant-length function.

3.1.2. Centered proximal-gradient update

Let $J_K = K^{-1}\mathbf{1}_K\mathbf{1}_K^\top$ and $H_K = I_K - J_K$. For a step size $s > 0$, set

$$v = \eta^{(m)} - s\nabla f_t(\eta^{(m)}).$$

The update separates by observed coordinate and has the closed form

$$\eta_{\cdot j}^{(m+1)} = J_K v_{\cdot j} + \left(1 - \frac{s\lambda_\eta w_j}{\|H_K v_{\cdot j}\|_2}\right)_+ H_K v_{\cdot j}. \quad (3)$$

The common component is therefore retained, while the centered contrast is group-thresholded; this is a proximal group soft-thresholding operation (Parikh and Boyd, 2014). Mixing proportions are updated by

$$\pi_k^{\text{cand}} = \frac{\max(N_k^{(t)}/n, 10^{-12})}{\sum_h \max(N_h^{(t)}/n, 10^{-12})}.$$

The initial step size is $s = 1/\max_k N_k^{(t)}$. If the smooth majorization condition

$$f_t(\eta^{(m+1)}) \leq f_t(\eta^{(m)}) + \langle \nabla f_t(\eta^{(m)}), \eta^{(m+1)} - \eta^{(m)} \rangle + \frac{\|\eta^{(m+1)} - \eta^{(m)}\|_F^2}{2s} \quad (4)$$

is not satisfied within numerical tolerance, s is halved and the proximal proposal is recomputed. This backtracking is repeated for at most 40 trials.

3.2. Guarded proximal generalized-EM algorithm

After the inner proximal-gradient update, the implementation evaluates the penalized auxiliary function and the observed penalized log-likelihood. The candidate is retained only if neither quantity decreases by more than 10^{-8} . Otherwise, the path fit stops, returns the previous accepted estimate, and records a failure diagnostic. Convergence is declared when the relative change in the observed penalized criterion is below 10^{-7} .

Components with effective count below 10^{-8} are retained to preserve the fixed value of K and the event is recorded. They are not pruned or reinitialized. These safeguards support statements about accepted numerical updates; they do not establish convergence of the complete parameter sequence, stationarity of every fitted path point, or global optimality.

Algorithm 1 summarizes the complete pathwise procedure. The same algorithm fits E-CGL and E-ACGL; only the fixed weights differ.

Algorithm 1. Guarded path algorithm for E-CGL and E-ACGL

Input: Data X , component number K , path size L , method indicator, iteration limits, and tolerances $\varepsilon_{\text{conv}}, \varepsilon_{\text{acc}}$.

Output: Selected support $\hat{S}$, its generating tuning set $\hat{\Lambda}_\eta$, representative $\hat{\lambda}_\eta = \max \hat{\Lambda}_\eta$; refitted parameters $\hat{\Theta}^{\text{refit}}$; numerical diagnostics.

Stage 1: Dense start and path construction

- 1: **Dense start:** fit the finite- κ dense model from multiple initializations and retain the converged fit with the largest observed log-likelihood.
- 2: **Weights:** set $w_j = 1$ for E-CGL; for E-ACGL, compute w_j from $\Theta^{(0)}$ and then hold w fixed.
- 3: **Path:** construct the KKT-geometric sequence $\Lambda = (0, \lambda_1, \dots, \lambda_{L-1})$ from dense to sparse.

Stage 2: Guarded penalized path

- 4: **for** $\lambda_\eta \in \Lambda$ **do**; warm-start from the preceding accepted path fit.
- 5: **repeat**
- 6: **E-step:** evaluate τ_{ik} by (2); set $N_k \leftarrow \sum_i \tau_{ik}$, $r_k \leftarrow \sum_i \tau_{ik} x_i$.
- 7: **M-step:** set $\pi_k^+ \leftarrow N_k/n$ and η^+ by (3).
- 8: **Step control:** recompute the candidate while halving s until (4) and both acceptance inequalities below hold, or the backtracking limit is reached.
- 9: **Acceptance:** retain Θ^+ if $\Delta Q_{\lambda_\eta} \geq -\varepsilon_{\text{acc}}$ and $\Delta \mathcal{L}_{\lambda_\eta} \geq -\varepsilon_{\text{acc}}$; after the backtracking limit, otherwise mark the path point as failed and terminate the path.
- 10: **until** $|\Delta \mathcal{L}_{\lambda_\eta}|/(1 + |\mathcal{L}_{\lambda_\eta}|) < \varepsilon_{\text{conv}}$ or the iteration limit is reached.
- 11: **Store:** S_{λ_η} , the fitted criterion, and diagnostics.
- 12: **end for**

Stage 3: Support-constrained refit and selection

- 13: **Refit:** for each distinct S , compute $\hat{\Theta}_S^{\text{refit}}$ subject to $H_K \eta_{\cdot j} = 0$ for $j \notin S$.
 - 14: **Score:** $\text{BIC}^{\text{refit}}(S) \leftarrow -2\ell(\hat{\Theta}_S^{\text{refit}}) + \log(n) \text{df}(S)$.
 - 15: **Select:** $\hat{S} \leftarrow \arg \min_S \text{BIC}^{\text{refit}}(S)$.
 - 16: **Return:** $\hat{S}$, $\hat{\Lambda}_\eta$, $\hat{\lambda}_\eta = \max \hat{\Lambda}_\eta$, $\hat{\Theta}^{\text{refit}}$, and diagnostics.
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3.3. Pathwise selection and support-constrained refit

3.3.1. KKT-geometric path

The path starts at $\lambda_\eta = 0$ and uses warm starts. Responsibilities from the unpenalized dense initializer define (N_k, r_k) . Let $r_\bullet = \sum_k r_k$ and $N_\bullet =$

$\sum_k N_k = n$, and let η^{common} be the pooled finite-cap vMF estimate formed from $(N_\bullet, r_\bullet)$. The matrix η_0 repeats η^{common} across components. A KKT-scale upper endpoint for this initial fixed-responsibility subproblem is

$$\lambda_{\max} = 1.05 \max_{1 \leq j \leq d} \frac{\|H_K \nabla f_t(\eta_0)_{\cdot j}\|_2}{w_j}.$$

The confirmatory path contains zero and 119 positive values on a logarithmic grid from $0.01\lambda_{\max}$ to $\lambda_{\max}$. It stops early if the active support has at most one coordinate. The principal simulations therefore use at most 120 path points.

3.3.2. Target-preserving support-constrained refit

The penalized path identifies candidate supports, but the final estimates are obtained by a sparsity-penalty-free, finite-cap constrained refit. For a candidate support S , impose

$$j \notin S \implies c_{\cdot j} = 0 \iff \eta_{1j} = \dots = \eta_{Kj} = \bar{\eta}_j.$$

Thus, an inactive posterior-score coordinate retains an estimated common baseline in the natural-parameter space. The constrained parameterization is

$$\eta_{kj} = b_j + c_{kj}, \quad \sum_{k=1}^K c_{kj} = 0, \quad c_{kj} = 0 \quad (j \notin S).$$

For each active coordinate, the first $K - 1$ contrasts are free and the last is set to their negative sum. Conditional on the responsibilities, the M-step maximizes

$$Q_S(\eta) = \sum_{k=1}^K [\eta_k^\top r_k + N_k \log C_d(\|\eta_k\|_2)]$$

over $b \in \mathbb{R}^d$ and the free active contrasts. The implementation uses gradient calculations consistent with the numerically evaluated vMF normalizer and L-BFGS-B, followed by step-halving if the observed log-likelihood decreases. Responsibilities, mixing proportions, and the constrained natural parameters are iterated until the numerical stopping criterion is met.

The target-preserving support-constrained refit enforces the centered support constraint directly while retaining the common baseline. The constrained likelihood is optimized numerically and is not claimed to attain the global maximum of the non-convex mixture likelihood.

The earlier active-only refit imposed $\eta_{\cdot j} = 0$ outside S and therefore removed both the centered contrast and the common baseline. It is retained only as an ablation diagnostic; the target-preserving support-constrained refit is used in the primary analyses.

3.3.3. Support selection by BIC after refit

Let $m_S = |S|$. The practical degrees-of-freedom approximation is

$$\text{df}(S) = d + (K - 1)m_S + (K - 1)\mathbf{1}(m_S > 0).$$

The three terms count the coordinate-wise common baselines, the active centered contrasts, and the mixing proportions when nonzero component contrasts are present. This is a model-selection approximation, not an exact effective degrees of freedom for a penalized non-regular mixture.

For each unique support on the path, define

$$\text{BIC}^{\text{refit}}(S) = -2\ell(\hat{\Theta}_S^{\text{refit}}) + \log(n) \text{df}(S).$$

The final support is

$$\hat{S} = \arg \min_{S \in \mathcal{S}_{\text{path}}} \text{BIC}^{\text{refit}}(S).$$

Several tuning values may generate the same support, so BIC after refit does not identify a unique tuning value. We report

$$\hat{\Lambda}_\eta = \{\lambda_\eta : S_{\lambda_\eta} = \hat{S}\}, \quad \hat{\lambda}_\eta = \max \hat{\Lambda}_\eta$$

as a deterministic representative convention.

All distinct supports encountered on the path are refitted. The information criterion follows the BIC principle of Schwarz (1978), using the practical degrees-of-freedom approximation above.

An EBIC sensitivity criterion is

$$\text{EBIC}_\gamma(S) = \text{BIC}^{\text{refit}}(S) + 2\gamma \log \left(\frac{d}{m_S} \right), \quad \gamma \in \{0.25, 0.5, 1\}.$$

EBIC (Chen and Chen, 2008) and alternative degrees-of-freedom rules are treated as sensitivity analyses rather than changes to the primary selector.

3.3.4. Selection of the number of components

Component-number selection is not part of the E-CGL support estimator. The number of components and the sparsity level are therefore selected in separate stages. In support-recovery simulations, K is fixed at the data-generating value so that support selection is not confounded with component-number misspecification.

For an unknown K , let $\mathcal{K}$ be a prespecified candidate set. The practical two-stage procedure is:

1. Fit an unpenalized dense vMF mixture for each $K \in \mathcal{K}$ using multiple starts.
2. Compare candidate K values using label-free density criteria, information criteria, and partition stability as appropriate to the application.
3. Fix the selected component resolution $\hat{K}$.
4. Fit the E-CGL path at $\hat{K}$ and choose a distinct support by BIC after the target-preserving support-constrained refit.

Symbolically,

$$\hat{K} = \arg \min_{K \in \mathcal{K}} \mathcal{C}_{\text{dense}}(K), \quad \hat{S}_{\hat{K}} = \arg \min_{S \in \mathcal{S}_{\text{path}}(\hat{K})} \text{BIC}^{\text{refit}}(\hat{K}, S).$$

Here $\mathcal{C}_{\text{dense}}$ may use independent or held-out negative log-likelihood, bootstrap out-of-bag negative log-likelihood, or an information criterion. No single information criterion is asserted to select the correct resolution in every high-dimensional mixture. When density fit and stability favor different K , both competing resolutions are reported.

3.3.5. Computational implementation

The implementation uses log-sum-exp stabilization in the E-step, warm starts along the penalty path, and explicit convergence and constraint checks in the centered-support refit. Optional Rcpp helpers replace low-level numerical operations without changing the statistical algorithm; R-only and Rcpp-helper outputs were checked to numerical tolerance before runtime comparisons.

The current computational claims are limited to empirical equality and runtime diagnostics. The method does not rely on an Rcpp implementation for its statistical definition.

The likelihood of a finite vMF mixture can be unbounded under component concentration collapse (Ng, 2023). We therefore use the same finite numerical parameter space in dense initialization, the penalized path, and support-constrained refitting:

$$0 \leq \kappa_k = \|\eta_k\|_2 \leq \texttt{kappa_cap} = 10^6.$$

Backtracking rejects any proposed E-CGL or E-ACGL iterate outside this space. A converged refit that reaches 99% of the bound is retained for diagnosis but is not eligible for information-criterion selection. This guard defines the finite computational problem; it is not an additional sparsity penalty. “Exact refit” refers to exact enforcement of the centered-support constraint, not to a global mixture optimum. The algorithmic results below do not claim convergence to a global maximizer.

3.4. Structural properties

The following statements justify the centered support estimand for a fixed number of components. They do not imply uniqueness of a fitted mixture, which remains subject to finite-mixture identifiability and label permutation. Full proofs are given in Appendix A.

Proposition 1 (Centered decomposition and pairwise support). *For any $v \in \mathbb{R}^K$, there is a unique decomposition*

$$v = \bar{v}\mathbf{1}_K + c, \quad \bar{v} = K^{-1}\mathbf{1}_K^\top v, \quad \mathbf{1}_K^\top c = 0.$$

For $v = \eta_j$, the following statements are equivalent:

$$c = 0; \quad \eta_{1j} = \cdots = \eta_{Kj}; \quad \eta_{kj} - \eta_{\ell j} = 0 \quad \text{for all } k < \ell.$$

Moreover,

$$\|c\|_2^2 = \frac{1}{K} \sum_{k < \ell} (\eta_{kj} - \eta_{\ell j})^2.$$

Consequently,

$$S_\eta = \bigcup_{k < \ell} \{j : \eta_{kj} - \eta_{\ell j} \neq 0\},$$

the union of the feature-dependent linear supports in all pairwise posterior log-odds.

Proposition 1 clarifies the scope of the estimand. If $c_j = 0$, the coefficient of x_j cancels from every pairwise posterior log-odds. The common baseline is nevertheless retained in η_k and may affect the intercept through $\|\eta_k\|_2$; it is therefore not irrelevant to the component density.

Proposition 2 (Label equivariance and support invariance). *Let P be a $K \times K$ permutation matrix. Then*

$$H_K P = P H_K, \quad \|H_K P \eta_j\|_2 = \|H_K \eta_j\|_2.$$

The E-CGL penalty is invariant to component relabeling, its proximal map is equivariant, and S_η is unchanged.

For a nonzero natural parameter, the component representation is recovered as $\kappa_k = \|\eta_k\|_2$ and $\mu_k = \eta_k / \|\eta_k\|_2$. At $\eta_k = 0$, the vMF component is uniform and its direction is not identified. These algebraic statements do not establish finite-mixture identifiability or uniqueness of the fitted mixture. Component-specific interpretation assumes distinct component distributions and positive mixing weights and is made only up to label permutation. If two components have identical natural parameters, their individual mixing weights are not separately identified. Support-recovery statements are conditional on fixed K .

3.5. Algorithmic properties

Proposition 3 (Closed-form proximal update). *Let $J_K = K^{-1} \mathbf{1}_K \mathbf{1}_K^\top$ and $H_K = I_K - J_K$. For $a \geq 0$ and $v \in \mathbb{R}^K$,*

$$\text{prox}_{a\|H_K \cdot\|_2}(v) = J_K v + \left(1 - \frac{a}{\|H_K v\|_2}\right)_+ H_K v,$$

where the second term is defined as zero when $H_K v = 0$.

The common baseline is unchanged. Group soft thresholding is applied only to the centered contrast, matching the coordinate-level definition of S_η .

Proposition 4 (Smooth majorization and proximal descent). *Fix responsibilities with $N_k > 0$ and fixed weights $w_j > 0$. The function f_t in Section 3.1.1 is convex and has a Lipschitz gradient with constant $L \leq \max_k N_k$. Hence, for $s \leq 1/L$, the proximal-gradient update in Section 3.1.2 does not increase F_t in exact arithmetic. More generally, an update accepted with majorization tolerance ε_{maj} satisfies*

$$F_t(\eta^{(m+1)}) \leq F_t(\eta^{(m)}) + \varepsilon_{\text{maj}}.$$

Proposition 4 motivates the backtracking rule. The implementation starts from $s = 1/\max_k N_k$ and halves the step until the majorization inequality is satisfied numerically. This check also protects against error from numerical evaluation of the vMF normalizing constant.

Let $Q(\Theta \mid \tau^{(t)})$ denote the expected complete-data log-likelihood under the current responsibilities, and define

$$Q_{\lambda_\eta}(\Theta \mid \tau^{(t)}) = Q(\Theta \mid \tau^{(t)}) - \lambda_\eta \sum_{j=1}^d w_j \|H_K \eta_j\|_2.$$

Proposition 5 (Penalized generalized-M monotonicity). *Let $\tau^{(t)}$ be the posterior responsibilities under $\Theta^{(t)}$. If a candidate $\Theta^{(t+1)}$ satisfies*

$$Q_{\lambda_\eta}(\Theta^{(t+1)} \mid \tau^{(t)}) \geq Q_{\lambda_\eta}(\Theta^{(t)} \mid \tau^{(t)}),$$

then, in exact arithmetic,

$$\mathcal{L}_{\lambda_\eta}(\Theta^{(t+1)}) \geq \mathcal{L}_{\lambda_\eta}(\Theta^{(t)}).$$

The implementation verifies both inequalities directly and permits decreases of at most 10^{-8} , so its finite trace is only numerically near-monotone. For the exact-arithmetic algorithm with zero acceptance tolerance, a bounded penalized criterion has a nondecreasing and therefore convergent sequence of objective values. These results do not establish boundedness or convergence of the full parameter sequence, stationary-point convergence, or global optimality.

4. Numerical studies

4.1. Design and comparison methods

The primary experiment evaluates recovery of posterior-score contrast support while varying sample size, component overlap, and concentration heterogeneity. Data are generated from

$$x_i \mid z_i = k \sim \text{vMF}(\mu_k, \kappa_k), \quad \eta_k = \kappa_k \mu_k, \quad P(z_i = k) = K^{-1}.$$

Table 1: Primary Study B design. Each of the 12 cells uses 100 repetitions.

quantity	values
components and dimension	$K = 4, d = 200$
sample size	$n \in \{300, 1000\}$
target oracle error	$e_B \in \{0.025, 0.05, 0.10\}$
equal concentration	$(45, 45, 45, 45)$
heterogeneous concentration	$(30, 40, 50, 60)$
support sizes	$(q_C, q_D, q_N) = (4, 16, 180)$
initialization and path	10 starts; at most 120 path points
support selector	BIC after target-specific support refit

The coordinates are partitioned as

$$\{1, \dots, d\} = S_C \cup S_D \cup S_N, \quad (q_C, q_D, q_N) = (4, 16, 180).$$

Coordinates in S_C have the same natural-parameter value in every component, coordinates in S_D carry centered component contrasts, and coordinates in S_N are zero. The 16 decision coordinates are divided into four equal blocks. For block g , the unscaled contrast is

$$v_{kg} = \begin{cases} 1, & k = g, \\ -1/(K-1), & k \neq g. \end{cases}$$

The component-specific scaling preserves the prescribed κ_k , and the common-to-decision signal ratio is calibrated by bisection to attain the target oracle Bayes error

$$e_B = P \left\{ \arg \max_{1 \leq k \leq K} P(z = k \mid X; \Theta) \neq Z \right\}.$$

Table 1 summarizes the primary simulation design.

An independent Monte Carlo sample of size 200,000 gave achieved oracle errors of 2.5035%, 5.0970%, and 9.8825% under equal concentrations and 2.6640%, 5.0685%, and 10.0365% under heterogeneous concentrations. The largest absolute calibration discrepancy was 0.164 percentage points.

Table 2 defines the six methods in the frozen Study B comparison. Dense vMF estimates unrestricted component-specific concentrations. M-L targets sparse direction entries and provides the sparse-vMF literature reference. M-GL and M-AGL are nonadaptive and adaptive direction-group diagnostics;

Table 2: Comparison methods and support targets in the primary simulation.

method	penalty or constraint	support target	role
Dense vMF	none	all coordinates	likelihood baseline
M-L	$\lambda_\mu \sum_{k,j} \mu_{kj} $	prototype entries	literature reference
M-GL	$\lambda_\mu \sum_j \ \mu_{\cdot j}\ _2$	prototype coordinates	group diagnostic
M-AGL	$\lambda_\mu \sum_j w_j^{(M)} \ \mu_{\cdot j}\ _2$	prototype coordinates	adaptive diagnostic
E-CGL	$\lambda_\eta \sum_j \ c_{\cdot j}\ _2$	posterior-score coordinates	primary method
E-ACGL	$\lambda_\eta \sum_j w_j^{(E)} \ c_{\cdot j}\ _2$	posterior-score coordinates	adaptive extension

they are not methods proposed by Rossi and Barbaro (2022). E-CGL is the primary method and E-ACGL is its adaptive extension. The matched centered entry-wise method E-CL is examined separately as a penalty-structure ablation in the Supplement.

Each repetition uses the same sample and dense initialization across methods. Each penalized method is evaluated after a refit that respects its own support target. For E-CGL and E-ACGL, every distinct path support is refitted under the exact constraint

$$j \notin S \implies c_{\cdot j} = 0, \quad \eta_{1j} = \dots = \eta_{Kj} = \bar{\eta}_j,$$

and is selected by the BIC in Section 3.3.3. Adaptive weights use $\gamma = 1$, $\epsilon = 10^{-6}$, and median normalization.

Clustering recovery is measured by adjusted Rand index (ARI). Support recovery is summarized by selected support size, true-positive and false-positive rates, precision, and $F_{1,\eta}$ relative to the true posterior-score contrast support $S_\eta = S_D$, together with

$$\hat{q}_C = |\hat{S} \cap S_C|, \quad \hat{q}_D = |\hat{S} \cap S_D|, \quad \hat{q}_N = |\hat{S} \cap S_N|.$$

After label alignment, centered natural-parameter error is

$$\text{MSE}_{\eta^c} = \frac{1}{Kd} \sum_{k=1}^K \sum_{j=1}^d (\hat{c}_{kj} - c_{kj})^2.$$

Table 3: Study B means at $n = 1000$, pooling the two concentration patterns. The true posterior-score support size is 16.

e_B	method	$\hat{q}$	$\hat{q}_C$	$\hat{q}_D$	$\hat{q}_N$	$F_{1,\eta}$	ARI	MSE $_{\eta^c}$
2.5%	Dense vMF	200.00	4.00	16.00	180.00	0.148	0.915	0.689
2.5%	M-L	200.00	4.00	16.00	180.00	0.148	0.915	0.687
2.5%	M-GL	20.00	4.00	16.00	0.00	0.889	0.930	0.068
2.5%	M-AGL	20.00	4.00	16.00	0.00	0.889	0.930	0.068
2.5%	E-CGL	16.01	0.00	16.00	0.01	1.000	0.930	0.054
2.5%	E-ACGL	16.01	0.00	16.00	0.01	1.000	0.930	0.054
5.0%	Dense vMF	200.00	4.00	16.00	180.00	0.148	0.831	0.745
5.0%	M-L	199.99	4.00	16.00	179.99	0.148	0.832	0.742
5.0%	M-GL	20.00	4.00	16.00	0.00	0.889	0.865	0.073
5.0%	M-AGL	20.00	4.00	16.00	0.00	0.889	0.865	0.073
5.0%	E-CGL	16.01	0.00	16.00	0.01	1.000	0.865	0.059
5.0%	E-ACGL	16.01	0.00	16.00	0.01	1.000	0.865	0.059
10.0%	Dense vMF	200.00	4.00	16.00	180.00	0.148	0.675	0.938
10.0%	M-L	199.99	4.00	16.00	179.99	0.148	0.677	0.929
10.0%	M-GL	19.63	4.00	15.44	0.19	0.854	0.719	0.363
10.0%	M-AGL	20.00	4.00	16.00	0.00	0.889	0.746	0.083
10.0%	E-CGL	16.16	0.02	16.00	0.15	0.995	0.747	0.070
10.0%	E-ACGL	16.03	0.01	16.00	0.02	0.999	0.747	0.068

Zero-support outcomes enter support summaries with $F_{1,\eta}$ equal to zero. For Dense vMF and the prototype-oriented M-L, M-GL, and M-AGL methods, $F_{1,\eta}$ is a cross-target evaluation against S_η , not an assessment of recovery of each method’s own support target.

4.2. Posterior-score contrast support recovery

Table 3 reports means over the equal- and heterogeneous- concentration cells with equal weight. The corresponding cellwise results are reported separately in Table S1; no equivalence claim is based on the pooled means. At $n = 1000$, E-CGL selected on average 16.01, 16.01, and 16.16 coordinates as e_B increased, with $F_{1,\eta}$ of 1.000, 1.000, and 0.995. It selected almost no common coordinates. M-GL and M-AGL generally recovered the posterior-score coordinates but retained the four common coordinates, reflecting their prototype-coordinate target. Dense vMF and M-L selected nearly all coordinates.

At $n = 300$, both E-series procedures remained close to the true support in the 2.5% and 5% oracle-error settings. The principal finite-sample degra-

Table 4: Study B means at $n = 300$ and target $e_B = 10\%$, pooling the two concentration patterns.

method	$\hat{q}$	$\hat{q}_C$	$\hat{q}_D$	$\hat{q}_N$	$F_{1,\eta}$	ARI	MSE_{η^c}
Dense vMF	200.00	4.00	16.00	180.00	0.148	0.477	4.382
M-L	199.99	4.00	16.00	179.99	0.148	0.488	4.328
M-GL	20.59	4.00	15.38	1.21	0.842	0.700	0.647
M-AGL	20.01	4.00	15.48	0.54	0.859	0.706	0.516
E-CGL	17.78	0.07	15.33	2.39	0.916	0.691	0.546
E-ACGL	15.83	0.01	15.46	0.36	0.971	0.721	0.356

dation occurred at $e_B = 10\%$ (Table 4). In the hardest heterogeneous cell, E-CGL selected 19.34 coordinates, including 14.66 decision and 4.56 noise coordinates, with $F_{1,\eta} = 0.838$ and ARI 0.651. E-ACGL reduced false positives in that cell and had $F_{1,\eta} = 0.951$ and ARI 0.712. This difference is specific to the difficult small-sample regime; E-CGL remains the primary estimator because its penalty does not depend on an initial adaptive weight.

Across the 12 primary cells, the maximum Monte Carlo standard errors over all methods (and over E-CGL/E-ACGL) were 0.0229 (0.0081) for $F_{1,\eta}$, 0.0194 (0.0064) for ARI, and 0.205 (0.047) for MSE_{η^c} . The Supplement describes their calculation and reports the full E-series cell means.

Figures 1, 2, and 3 show the repetition-level distributions. They use the same 7,200 frozen method-repetition rows as the tables and display sample size and concentration pattern in separate panels.

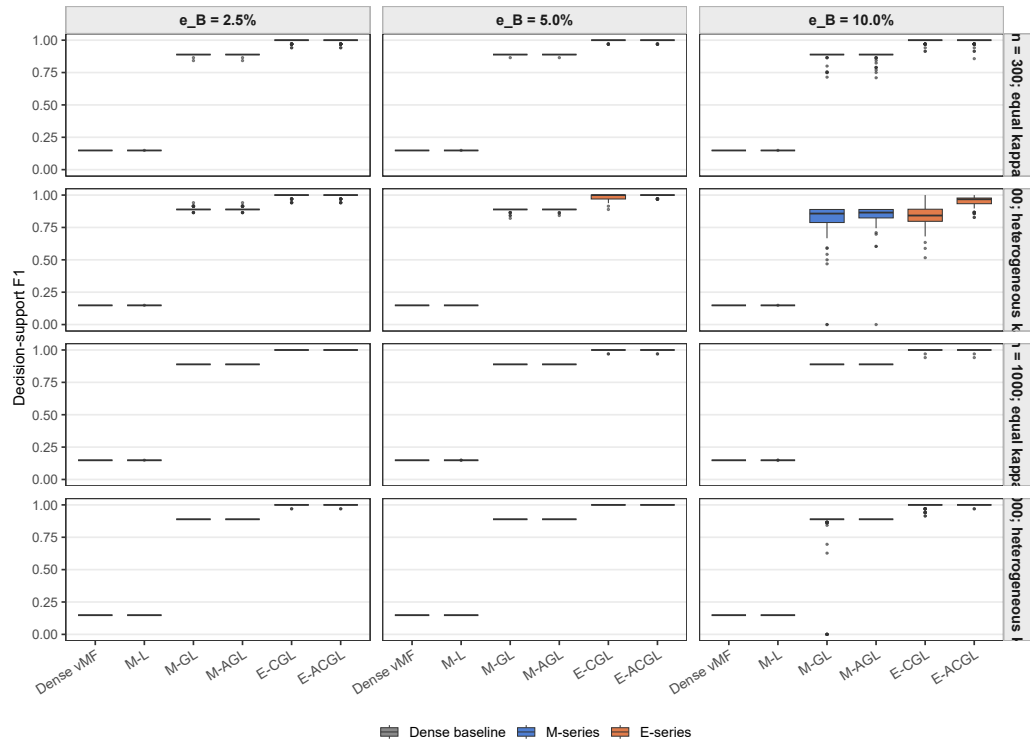


Figure 1: Posterior-score contrast-support $F_{1,\eta}$ across Study B cells.

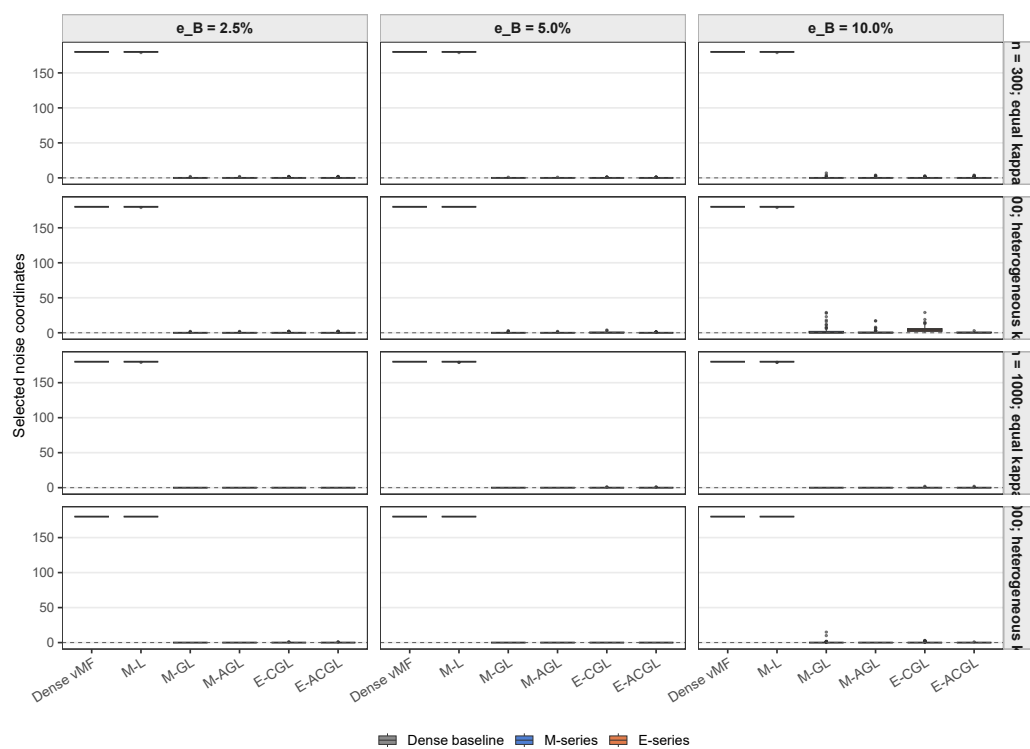


Figure 2: Number of selected noise coordinates across Study B cells.

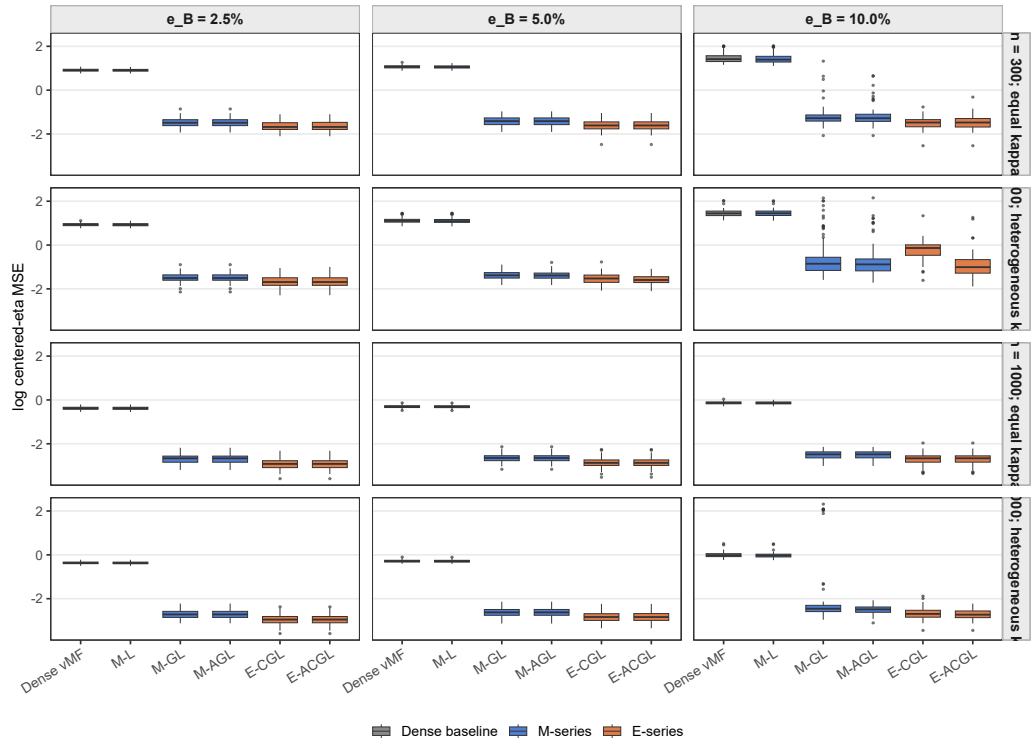


Figure 3: Log centered natural-parameter MSE across Study B cells.

Table 5: Structural-scope experiments. $F_{1,\eta}$ is defined relative to posterior-score contrast support. The dense-support rows contrast small and large samples under equal concentrations.

design	method	$\hat{q}$	$\hat{q}_C$	$\hat{q}_D$	$\hat{q}_N$	$F_{1,\eta}$	ARI
shared background	Dense vMF	200.00	80.00	20.00	100.00	0.182	0.833
shared background	M-L	199.94	80.00	20.00	99.94	0.182	0.833
shared background	M-GL	35.26	15.26	20.00	0.00	0.766	0.865
shared background	M-AGL	41.18	21.16	20.00	0.02	0.695	0.861
shared background	E-CGL	20.02	0.00	20.00	0.02	1.000	0.868
shared background	E-ACGL	20.02	0.00	20.00	0.02	1.000	0.868
dense, $n = 300$	Dense vMF	200.00	4.00	80.00	116.00	0.571	0.419
dense, $n = 300$	M-L	199.98	4.00	80.00	115.98	0.571	0.429
dense, $n = 300$	M-GL	4.08	4.00	0.06	0.02	0.001	0.001
dense, $n = 300$	M-AGL	4.08	4.00	0.06	0.02	0.001	0.001
dense, $n = 300$	E-CGL	22.68	0.16	17.70	4.82	0.244	0.134
dense, $n = 300$	E-ACGL	23.38	0.06	21.78	1.54	0.363	0.198
dense, $n = 1000$	Dense vMF	200.00	4.00	80.00	116.00	0.571	0.681
dense, $n = 1000$	M-L	199.98	4.00	80.00	115.98	0.571	0.683
dense, $n = 1000$	M-GL	4.02	4.00	0.02	0.00	0.000	0.000
dense, $n = 1000$	M-AGL	47.62	4.00	43.18	0.44	0.524	0.389
dense, $n = 1000$	E-CGL	79.96	0.02	79.50	0.44	0.994	0.719
dense, $n = 1000$	E-ACGL	78.92	0.00	78.86	0.06	0.992	0.717

4.3. Structural scope of the support target

Two prespecified experiments examine settings in which prototype and decision support differ or in which the sparse posterior-score-support assumption is weakened. The shared-background design uses $(q_C, q_D, q_N) = (80, 20, 100)$, $n = 1000$, heterogeneous concentrations, and $e_B = 5\%$. The dense-support design uses $(4, 80, 116)$, equal concentrations, and $e_B = 10\%$ at $n = 300$ and $n = 1000$. All cells use 50 repetitions and the same target-specific refit rules as Study B.

Table 5 reports the resulting support decomposition.

In the shared-background design, E-CGL removed the 80 common coordinates while retaining all 20 posterior-score coordinates. The prototype-oriented M-GL and M-AGL methods retained part of the common background because their estimand is prototype-coordinate support. In the dense-support design at $n = 300$, E-CGL selected only 17.70 of 80 posterior-score coordinates and had ARI 0.134, whereas Dense vMF and M-L had ARI 0.419 and 0.429 without sparse support recovery. At $n = 1000$, E-CGL

Table 6: Proportion selecting the true $K = 4$ in the dense-model diagnostic.

criterion	equal κ	heterogeneous κ
AIC	1.00	1.00
BIC	0.78	0.00
ICL-BIC	0.78	0.00
EBIC _{0.5}	0.00	0.00
RICc	0.00	0.00
independent test NLL	1.00	1.00

recovered 79.50 posterior-score coordinates with $F_{1,\eta} = 0.994$. Thus, dense weak posterior-score support can lead to severe finite-sample under-selection, although density alone does not preclude recovery when the sample is sufficiently large. The complete four-cell dense-support results are reported in the Supplement.

4.4. Selection of the number of components

Support recovery in the preceding experiments conditions on $K = K^* = 4$. A separate diagnostic fit dense vMF models for $K \in \{2, \dots, 8\}$ at $n = 1000$, $d = 200$, and $e_B = 5\%$. Table 6 reports the frequency with which each rule selected $K = 4$ over 50 repetitions. The independent test sample was used only to diagnose density selection and is not an available real-data selector.

In a separate 10-repetition resampling diagnostic, five-split holdout NLL selected $K = 4$ in 10/10 repetitions under both concentration patterns; its 1-SE rule selected $K = 4$ in 10/10 equal and 9/10 heterogeneous repetitions. These results motivate the two-stage procedure in Section 3.3.4: choose a component resolution using density prediction and stability diagnostics, then select posterior-score support by BIC after support-constrained refit. They do not identify a universally valid information criterion for K .

The frozen Study B output contains all 7,200 planned rows with no error rows; two standard-method fits reported nonconvergence. E-CGL and E-ACGL selected fits satisfied the majorization and line-search acceptance checks. The Supplement reports full E-series cell means, the matched penalty ablation, selector and degrees-of-freedom sensitivity, ambient-dimension experiments, the Rossi-style bridge, runtime, and selected-fit numerical checks.

5. Real data analysis

5.1. Datasets and locked protocol

Classic3 contains abstracts from the CISI, CRAN, and MED collections and has been used in directional document clustering (Banerjee et al., 2005; Greene and Cunningham, 2006). BBCSport contains articles on athletics, cricket, football, rugby, and tennis (Greene and Cunningham, 2006). After removing exact and near duplicates before splitting, the analyses contained 3,883 Classic3 documents and 704 BBCSport documents. Each dataset used five locked stratified 80/20 splits. The resulting train/test sizes were 3,105/778 for Classic3 and 561/143 for BBCSport.

Documents were encoded by a vocabulary-aligned SPLADE CoCondenser model at a fixed model revision (Formal et al., 2021); the exact model identifier and revision are recorded in the reproducibility material. For each split, vocabulary coordinates were ranked using training-set variance only. The top 2,000 coordinates were retained for Classic3 and the top 500 for BBCSport, after which every document vector was normalized to unit Euclidean norm. No test document had maximum cosine similarity of at least 0.98 with a training document.

The benchmark class count was fixed during fitting: $K = 3$ for Classic3 and $K = 5$ for BBCSport. Supplied labels were used only for stratification, verification of K , post-fit ARI/NMI evaluation, and post-fit component naming. They were not used for vocabulary ranking, initialization, parameter estimation, or support selection.

5.2. Methods and evaluation

The comparison included spherical k -means, dense vMF mixtures with shared and component-specific concentration, the Rossi-style M-L prototype selector, M-CGL, M-ACGL, E-CGL, and E-ACGL. All fits used 30 dense random starts. Centered M/E paths contained 240 candidates, and the primary M-L path allowed 600 updates. Each sparse method used its own target-preserving support-constrained refit, followed by a practical BIC based on the training observed log-likelihood and the method-specific nominal dimension. The centered directional objectives, refit constraints, and nominal dimensions for M-CGL and M-ACGL are given in Supplement Section S9.

Selected q has a target-specific interpretation: prototype-union support for M-L, centered directional support for M-CGL/M-ACGL, and centered

Table 7: Locked repeated-holdout results. Entries are mean (SD) over five splits. NLL/doc is held-out negative log-likelihood per document; lower is better. Selected q is interpreted relative to each method’s coordinate target.

Method	ARI	NMI	Selected q	q/d	NLL/doc
<i>Panel A: Classic3, $d = 2,000$, $K = 3$</i>					
Spherical k -means	0.970 (0.007)	0.946 (0.010)	2,000.0	1.000	—
Dense vMF, shared κ	0.970 (0.007)	0.946 (0.010)	2,000.0	1.000	-4,872.890
Dense vMF, free κ_k	0.973 (0.006)	0.953 (0.011)	2,000.0	1.000	-4,873.968
M-L	0.970 (0.009)	0.947 (0.013)	1,924.6	0.962	-4,872.239
M-CGL	0.970 (0.007)	0.949 (0.011)	1,376.8	0.688	-4,873.337
M-ACGL	0.971 (0.007)	0.950 (0.011)	1,345.4	0.673	-4,873.298
E-CGL	0.970 (0.007)	0.947 (0.012)	1,343.0	0.671	-4,873.297
E-ACGL	0.973 (0.006)	0.953 (0.011)	2,000.0	1.000	-4,873.967
<i>Panel B: BBCSport, $d = 500$, $K = 5$</i>					
Spherical k -means	0.894 (0.022)	0.897 (0.021)	500.0	1.000	—
Dense vMF, shared κ	0.898 (0.024)	0.901 (0.020)	500.0	1.000	-922.870
Dense vMF, free κ_k	0.877 (0.048)	0.878 (0.033)	500.0	1.000	-922.622
M-L	0.907 (0.020)	0.907 (0.019)	498.2	0.996	-921.605
M-CGL	0.880 (0.050)	0.882 (0.035)	303.2	0.606	-921.086
M-ACGL	0.880 (0.050)	0.882 (0.035)	302.4	0.605	-920.979
E-CGL	0.875 (0.053)	0.874 (0.042)	308.6	0.617	-921.211
E-ACGL	0.877 (0.048)	0.878 (0.033)	285.2	0.570	-920.796

posterior-score support for E-CGL/E-ACGL. Sparse-minus-dense comparisons were therefore paired with the matching concentration structure: shared κ for M-L and free κ_k for the centered methods. The five holdouts overlap, so the reported means, standard deviations, and split counts are descriptive rather than independent-replicate inference.

5.3. Classic3 interpretation

In Classic3, E-CGL excluded 32.9% of the coordinates from the estimated posterior-score contrast support while retaining their common natural-parameter baselines. Relative to dense free- κ_k vMF, its mean paired ARI difference was -0.0036 , and its held-out NLL increase was 0.6705 per document. ARI was lower in four splits and tied in one; NLL was higher in all five splits. Conditional on coordinates available in both training vocabularies, mean support Jaccard was 0.933 . E-ACGL selected the dense support in every split and is retained as an adaptive sensitivity result.

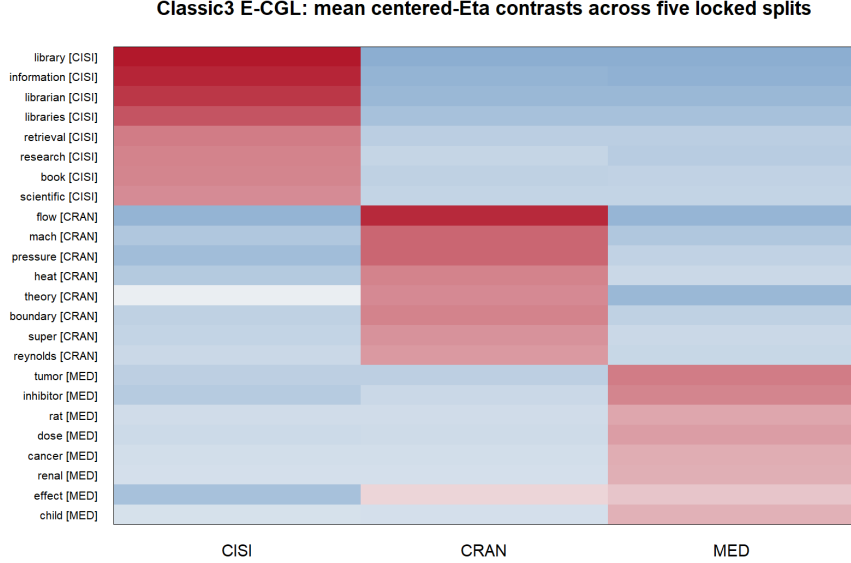


Figure 4: Classic3 E-CGL centered natural-parameter contrasts from the locked analysis. Labels were used only to name components after fitting.

After fitting, components were aligned to benchmark classes only for naming and visualization. Tokens selected in all five splits included *library*, *information*, and *retrieval* for CISI; *flow*, *pressure*, and *heat* for CRAN; and *tumor*, *inhibitor*, and *cancer* for MED.

5.4. BBCSport contrast case and numerical checks

In BBCSport, E-CGL excluded 38.3% of the coordinates from the estimated posterior-score contrast support. Its paired mean differences from dense free- κ_k vMF were -0.0026 for ARI and 1.4105 for NLL/doc. Every sparse method had higher held-out NLL than its matched dense model in all five splits. Raw cross-family ranks are not a penalty-only comparison because M-L used shared κ , whereas centered M/E methods used free κ_k .

All 80 method-split fits completed without error or selected-fit nonconvergence, and no selected support occurred at a path endpoint. BBCSport M-L stopped after a subsequent stronger-penalty fit failed in every split. A denser sensitivity path increased mean evaluated path rows from 331.8 to 771.2; selected q changed by at most one coordinate, and held-out ARI/NMI

were unchanged. The last evaluated support had BIC at least 19,497.94 larger than the selected support. Thus the near-dense M-L result is stable to the examined path resolution, although the later path failure remains a numerical limitation.

Classic3 M-CGL/M-ACGL produced high-order Bessel precision warnings. Over the selected-fit range, $d = 2,000$ and $\kappa \in [607, 818]$, the maximum relative difference between the production calculation and the high-dimensional reference was 2.850×10^{-10} , below the prespecified 10^{-6} tolerance. This is a range-specific audit, not a general error bound.

The CSTR analysis is retained only as a literature bridge: the implementation reproduced the published dense shared- κ and Rossi M-L ARI values. Its earlier E-series diagnostic used a different selector and is not combined with the locked BIC-after results. Real data supply no true coordinate support, so support TPR, FPR, precision, and F1 are not reported.

6. Discussion

6.1. Posterior-score contrast support as the estimand

This study defines posterior-score contrast support as the coordinates entering the feature-dependent linear term of pairwise posterior log-score contrasts. E-CGL regularizes centered natural-parameter contrasts and therefore targets posterior-score coordinates. The M-L, M-GL, and M-AGL methods target prototypes. The supplementary M-CGL and M-ACGL diagnostic targets centered directional support. Centering leaves the common natural-parameter baseline unpenalized, so a coordinate shared across components is not selected solely because it remains nonzero in every prototype. The coordinate-group penalty is justified by alignment between the regularization unit and this estimand, rather than by uniform empirical superiority over entry-wise regularization or prototype-oriented penalties.

6.2. Empirical scope and adaptive extension

Across the evaluated $n = 1000$ Study B cells, E-CGL selected 16.01–16.16 coordinates on average and attained $F_{1,\eta}$ between 0.995 and 1.000 for a true support size of 16 (Table 3). Clustering ARI decreased with increasing oracle Bayes error, whereas support recovery changed little in these cells; clustering and support difficulty therefore need not vary in parallel in the evaluated settings. The shared-background experiment illustrated the practical difference between prototype and posterior-score contrast support. The

dense-support experiment, in contrast, documented substantial finite-sample under-selection when decision information was distributed across many weak coordinates, although recovery improved at $n = 1000$ in that design.

E-ACGL improved false-positive control in some difficult small-sample cells but was otherwise similar to E-CGL (Table 4). We therefore retain E-CGL as the primary estimator and treat E-ACGL as an adaptive extension whose performance depends on the initial contrast estimates. In the final-engine matched ablation, the three centered penalties recovered the same support in the well-separated cell. In the difficult small-sample cell, E-CGL reduced noise selection relative to E-CL, and E-ACGL reduced it further. This result supports coordinate grouping in that regime but does not establish uniform superiority; the primary motivation remains the coordinate-level definition of the support target.

A supplementary target-separation diagnostic evaluated M-CGL and E-CGL against centered directional and centered natural-parameter support, respectively. Both methods recovered their targets in the common- concentration and common-natural-parameter designs. In the pure-concentration design, E-CGL recovered the nonempty canonical support with mean target F1 0.989, whereas M-CGL selected the correct empty directional support in 9 of 20 repetitions. The latter result documents path and numerical sensitivity at the empty-support boundary and is one reason M-CGL is retained as a directional diagnostic rather than a co-primary estimator.

The locked real-data analyses provided complementary evidence through held-out clustering and support stability rather than recovery of an unknown true support. In Classic3, E-CGL excluded 32.9% of the coordinates, with mean ARI and NLL costs of 0.0036 and 0.6705 per document relative to dense free- κ_k vMF (Table 7). Its conditional support Jaccard was 0.933, and the signed token contrasts were coherent across the five splits. In BBCSport, every sparse method had higher held-out NLL than its matched dense model in all five splits. CSTR is retained only as an implementation bridge for the Rossi-style method because its earlier E-series diagnostic used a different selector. These results delimit the intended use of E-CGL rather than establish a uniform ordering of support targets.

6.3. Model selection and computation

The support-recovery claims condition on a fixed number of components. The separate two-stage K diagnostic in Section 3.3.4 is a practical procedure

and does not establish consistency of component-number selection. In Classic3, held-out density favored finer component resolutions, whereas partition stability was maximized at the three supplied broad topics. The number of fitted density components and the resolution of an external label system should therefore be reported separately.

Support selection uses the unpenalized observed likelihood after exact centered-support refitting (Section 3.3.3). The corresponding BIC degrees of freedom are a practical model-selection approximation, not an exact effective-degrees-of-freedom result for a nonregular penalized mixture. Sensitivity analyses showed limited variation among the stored exact-refit candidates under the examined df and EBIC rules, but do not provide a general selection-consistency result.

The fitting algorithm controls accepted generalized-M updates through smooth majorization and backtracking, with a separate step-halving safeguard during exact refitting. These results concern objective control for accepted steps; they do not imply convergence of the full parameter sequence, stationarity, or global optimality. Optional Rcpp helpers reproduced the R-only outputs to numerical tolerance and reduced elapsed time in diagnostic benchmarks. This is an implementation result rather than a statistical performance claim.

6.4. *Limitations and future directions*

E-CGL is intended for settings in which posterior-score contrast support is sparse or compressible. Weak separation, many weak active coordinates, or small samples can produce under-selected or empty supports, and mixture fitting remains sensitive to initialization and local optima. The main simulations fix the true K to isolate support recovery, while the separate K experiments provide diagnostics rather than an inferential guarantee. Real-data support cannot be evaluated by TPR or F1 because its true coordinate support is unknown; the evidence instead rests on stability, held-out performance, and substantive interpretation. The Classic3 and BBCSport conclusions are also conditional on the fixed pretrained SPLADE revision and training-ranked vocabulary filters. The selected support is coordinate-system dependent: token-level interpretation is meaningful because the representation is aligned with the vocabulary. It does not transfer unchanged to an arbitrarily rotated dense embedding space.

Further work is needed on effective degrees of freedom for centered penalized mixtures, joint uncertainty in K and support, and selection rules that

remain stable under diffuse weak signals. Extensions that combine sparse decision support with explicit dense-background components may also reduce the finite-sample cost observed in dense-support settings.

Centered natural-parameter group regularization defines an estimator of sparse posterior-score contrast support in vMF mixtures. Its target differs from a sparse component prototype: it retains coordinates that change posterior component comparisons while leaving common natural-parameter baselines unpenalized. The simulation and Classic3 analyses support this target-specific interpretation, and the dense-support and supplementary real-data diagnostics identify settings in which sparsification can be costly. E-CGL is therefore positioned as a posterior-score-support estimator for sparse or compressible contrast structures, not as a uniformly superior replacement for dense or prototype-oriented mixture methods.

Appendix A. Proofs of propositions

Proof of Proposition 1. Taking the average of $v = \bar{v}\mathbf{1}_K + c$ and using $\mathbf{1}_K^\top c = 0$ determines $\bar{v}$ and hence c uniquely. The three zero-contrast statements are immediate. Finally, $\sum_{k < \ell} (v_k - v_\ell)^2 = K \sum_k (v_k - \bar{v})^2$, which gives the stated identity. $\square$

Proof of Proposition 2. Because $P\mathbf{1}_K = \mathbf{1}_K$, P commutes with J_K and H_K . Permutation matrices are orthogonal and therefore preserve the Euclidean norm. Applying Proposition 3 gives equivariance of the proximal map. $\square$

Proof of Proposition 3. The spaces spanned by $\mathbf{1}_K$ and orthogonal to $\mathbf{1}_K$ are orthogonal. The penalty is zero on the first space and equals the Euclidean norm on the second. The minimization therefore preserves $J_K v$ and applies the standard group soft-thresholding map to $H_K v$. $\square$

Proof of Proposition 4. The vMF log-partition function has Hessian equal to the covariance matrix of a unit-norm random vector and is therefore positive semidefinite with operator norm at most one. The Hessian of the k th block of f_t consequently has operator norm at most N_k . The descent result follows by combining the smooth majorization inequality with optimality of the proximal update. $\square$

Proof of Proposition 5. The standard EM lower-bound identity expresses the observed log-likelihood difference as the auxiliary-function difference plus a

nonnegative Kullback–Leibler divergence term. The penalty depends only on the parameters and can be subtracted from both sides, yielding the penalized result. $\square$

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